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Bhavitha Thummunuru
Praveen Kumar Mallireddy
Nikhil Reddy Gaddam

Student ID: 1. 25205692
2. 25211048
3. 25152190
4. 25169572
Programme: MSc in Artificial Intelligence Year: 2026
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Lecturer: Sallar Khan
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Date: 3rd April 2026
Project Title: AAn Agentic Structured-Data RAG System for Irish Housing Market Intelligence

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Praveen Kumar Mallireddy
Nikhil Reddy Gaddam

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Foundations of Artificial Intelligence
2nd April 2026

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AI Acknowledgment
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Perplexity was utilized to identify the academic research papers, articles, and journals related to House Rentals, for Foundations of AI. Used in understanding the formatting standards of IEEE reports. However, all written content was independently created and developed by the student.
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Additional Evidence:
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An Agentic Structured-Data RAG System for Irish Housing Market Intelligence
Foundation in AI
National College of Ireland
Student ID: 25205692
Bhavitha T
Foundation in AI
National College of Ireland
Student ID: 25211048
Praveen Kumar M
Foundation in AI
National College of Ireland
Student ID: 25152190
Nikhil Reddy Gaddam
Foundation in AI

National College of Ireland
Student ID: 25169572

Abstract
—Users must navigate multiple disconnected government datasets to understand Ireland's housing crisis. In this paper, we introduce a novel conversational tool designed to answer everyday questions about local rental and property markets. The system relies on a somewhat non-traditional Agentic Structured-Data Retrieval-Augmented Generation (ASD-RAG) approach. The system eliminates vector similarity search entirely. Instead, user queries are routed through a five-stage LangGraph pipeline that runs deterministic Pandas logic directly against official datasets (from the RTB, PPR, and CSO). While a modular Large Language Model—like LLaMA 3.3-70B or Google Gemini—takes care of reasoning and classification, Python strictly handles all the math. The system achieves sub-3-second latency with dynamic visualizations and exact source citations. We also implemented strict domain guardrails to refuse or redirect off-topic questions. Based on our empirical evaluation covering seven categories of intent, we observed a 99.1% numerical accuracy rate. For context, an unaided LLM recorded only 38.2%.
Index Terms—Retrieval-Augmented Generation, Multi-Agent Systems, LangGraph, Irish Housing Market, Conversational AI, FastAPI, Model-Agnostic LLM, Structured Data Retrieval, Affordability Analytics

I. Introduction

The Irish rental market continues to face severe strain. According to Q4 2024 figures from the RTB, the national average for new rents reached €1,680 per month. That represents a 5.5% annual increase, amounting to a 78% rise since 2015 [1]. At the same time, the CSO reported an 8.7% jump in its December 2024 Residential Property Price Index, underscoring the ongoing pressure on buyers [12]. In Dublin, premium market rents are increasingly challenging to access on a median wage [2].

Such data is technically public. However, it sits scattered across separate platforms, each requiring a different level of technical literacy to parse. The Property Price Register provides raw, unaggregated transaction logs, while the CSO's own portals demand familiarity with complex statistical dashboards. Navigating these disparate systems creates significant friction for the average renter or prospective buyer trying to cross-reference market prices against their personal income. Research by Hagedorn et al. [3] indicated that centralised platforms can reduce search times by 60–70%. Despite this proven need for accessible aggregation, a unified tool pairing Ireland's official housing data with natural-language questioning has previously not existed.

Large language models (LLMs) might seem like an obvious fix. But using one out of the box introduces serious risks. Models tend to invent numbers. Also, they lack any knowledge of events occurring after they were trained. Retrieval-Augmented Generation (RAG) was introduced by Lewis et al. [4] to fix exactly this by fetching external evidence to ground the model. Standard RAG setups rely on embedding similarity. This works fine for paragraphs of text, but performs poorly when dealing with tables where exact numeric filtering is the actual goal. Because of this limitation, we propose Agentic Structured-Data RAG (ASD-RAG). Here, we combine the reasoning capabilities of an LLM with deterministic data operations instead of guessing with vectors.

We present four main contributions:
1. A Conversational Interface: A functioning natural-language system covering official Irish housing data from 2010 to 2024.
2. Auditable Pipeline: A five-stage LangGraph system that explicitly separates intent, data retrieval, arithmetic, reasoning, and formatting.
3. Empirical Benchmarks: Our internal evaluations cover end-to-end latency, classification effectiveness, and an ablation study targeting the ASD-RAG architecture.
4. Decoupled Architecture: Agent logic is separated from the underlying LLM provider, allowing developers to switch between Gemini, LLaMA, or others seamlessly.

Following this section, we review relevant literature. Section III details our system architecture, while Section IV outlines the experimental results. Finally, limitations and next steps are covered in Section V.

II. Related Work

A. Retrieval-Augmented Generation

RAG was originally proposed by Lewis et al. [4] as a way to combine sequence-to-sequence models with an external index. It performed very well on standard QA benchmarks. That said, it relies heavily on the assumption that unstructured text can be fetched by similarity. When it comes to tabular filtering, this assumption breaks down.

Gao et al. [5] highlighted structured-data retrieval as a complex, open problem. They pointed out that simply flattening tables into text ruins numerical reasoning. TableRAG [6] attempts to solve this by alternating between text retrieval and SQL. With our proposed system, we took this idea further by eliminating approximate retrieval entirely. Every piece of data is accessed via hard-coded Pandas operations, ensuring every numeric claim maps back to a specific spreadsheet row.

B. Multi-Agent LLM Orchestration

Workflows can be modeled as stateful directed graphs, which is exactly what LangGraph does [7]. Nodes act as distinct agents, passing shared state back and forth through edges. Single-pass prompting often struggles with complex logic. In contrast, graph-based structures allow us to build pipelines where stages can be independently audited. As Wang et al. [8] discovered, using specialized sub-agents usually leads to better performance on multi-step queries than treating the LLM as a monolith. We relied on these findings when designing our own agent decomposition strategy.

Answering off-topic questions badly can erode user trust very quickly. Shneiderman’s framework for Human-Centred AI [9] suggests that these assistants need to support decision-making rather than act as a substitute for it. Transparency and tight focus are both required. To achieve this, the system implements a firm system-prompt guardrail inside the reasoning agent, blocking any input not related to mortgages, rentals, or housing policy.

D. Irish Housing Data and Information Asymmetry

Information asymmetry acts as a structural barrier in the housing market. Norris and Byrne [10] explained how eligible renters routinely miss out on support schemes simply because the required details are difficult to locate. Looking at European social housing, Scanlon et al. [11] documented similar issues, noting that centralized access platforms noticeably increased scheme participation. Our architecture was built to address exactly this gap. By fusing RTB, CSO, and PPR data, it offers a

single entry point for resources otherwise split across different portals with varying update schedules.

E. Agentic AI for Structured-Data QA

In their work on ReAct, Yao et al. demonstrated how interleaving tool calls with reasoning traces improves factual accuracy. The proposed ASD-RAG model applies a similar concept to structured data. Instead of browsing the internet, our agents use Pandas operations to pull exact statistics. Furthermore, our architecture demonstrates that when LLMs are fed structured numbers, they provide much richer contextual explanations compared to when they are forced to recall figures from memory.

III. System Architecture and Methodology

A. Architectural Overview

At its core, the backend operates as a FastAPI service resting on a single /api/chat endpoint. Incoming requests enter a compiled LangGraph state machine. This machine consists of five sequential nodes: Intent, Retrieval, Analysis, Reasoning, and Response. The output is a formatted JSON object carrying the answer, dynamic chart data, and the relevant sources. We serve the frontend as static files from this exact same process.

Fig. 1. Architectural diagram of the LangGraph pipeline. Purple nodes indicate LLM agents; green nodes represent deterministic operations.

The state graph compiles at startup and persists. A HousingState TypedDict holds all intermediate data. Because of this, both retrieval and analysis are completely deterministic for any given query. The LLM reasoning step is the only place variation is introduced. A normal round-trip takes less than three seconds.

B. Data Layer

We rely on three primary government datasets to ground the system’s responses. The RTB rent index file is roughly 5.7 MB, tracking precise county-level and local electoral area rental histories from 2010 to 2024. Next, the Dublin subset of the Property Price Register (around 27 MB) logs individual property sale transactions, documenting the exact addresses, dates, and final execution prices. Finally, the CSO Master Dataset provides localized median earnings and county-level mean sale prices, serving as the foundational macroeconomic baseline that allows our affordability agent to accurately calculate price-to-earnings ratios and local housing cost burdens.

To save memory, all files load lazily using Python’s lru_cache. Each CSV is read from disk only once. Upon loading, county names get title-cased and dates are parsed day-first. The data_manager exposes typed functions (such as get_rent_stats or compare_counties) that return dictionaries matching a fixed schema. Because of this strict schema, the underlying CSV files could easily be swapped out for a SQL database later without altering the agent logic.

C. Intent Agent

First up is the Intent Agent. Operating at a low temperature of 0.1, it calls the designated LLM API to classify the user’s goal into one of seven distinct categories, such as rent_query or affordability. We kept the temperature low because we need structural consistency, not creative writing. It also pulls out necessary entities like the year or target county.

can normalize messy inputs automatically. If it happens to generate invalid JSON, a regex fallback assigns a default general_chat intent. This prevents the whole pipeline from crashing over a formatting glitch.

D. Retrieval Agent

Once the intent is known, the Retrieval Agent triggers the proper data_manager calls. If someone asks for rent stats, it evaluates the primary numbers and then attempts to fetch a five-year trend using a try/except block. For single-county comparisons, the national average is grabbed automatically as a baseline.

Notice that no vector embedding happens here. We rely entirely on Pandas boolean masks running across in-memory DataFrames. Every statistic passed to the LLM traces back to a specific row, meeting the auditability standards recommended by Norris and Byrne [10].

E. Analysis Agent

The Analysis Agent takes the raw retrieval data and performs rigid, domain-specific macroeconomic calculations. Rent queries receive weekly cost breakdowns alongside calculated gross income requirements based on the standard 30% affordability rule, which posits that households should not allocate more than one-third of their gross income to secure adequate housing. House price questions trigger mortgage estimators, projecting monthly repayment schedules based on current prevailing interest rates, typical 30-year terms, and standard minimum deposit regulations. We calculate rolling multi-year trend trajectories directly to explicitly flag rapidly appreciating or depreciating local markets. Finally, we assign an overall affordability rating based on a five-tier scale tied tightly to corresponding local median earnings.

These math operations bypass the LLM completely. We store the resulting summary inside HousingState, preparing it for the narrative reasoning step coming next.

F. Reasoning Agent

Running at a higher temperature of 0.4, the Reasoning Agent turns the calculated data into readable text. The prompt treats the LLM as an expert Irish housing analyst. We deliberately included instructions on Rent Pressure Zones and Housing Assistance Payments. Crucially, a hard guardrail instructs the model to refuse off-topic inquiries.

Because the model receives pre-calculated math and strict context dictionaries, its job is simply to explain what the numbers mean rather than invent them. Responses are limited to roughly 300 words and heavily utilize bullet points to present information clearly.

G. Response Agent

Our final node, the Response Agent, is responsible for packaging the diverse intermediate outputs into a cohesive JSON payload ready for direct frontend ingestion. It carefully formats the narrative reasoning block, constructs dynamic Chart.js configuration objects—including setting correct geometric axes, responsive tooltips, and appropriate data labels based on the query context—flattens critical numerical metrics for immediate display, and strictly logs the exact dataset sources to guarantee traceability. We mapped specific chart types to specific analytical intents. For instance, multi-county comparisons might trigger a clean radar chart, whereas historical cost trends produce an intuitive multi-series line graph. We purposefully injected a small element of controlled randomization into the chart selection process to ensure the user interface remains visually engaging and does not feel monotonous over extended conversational sessions.

IV. Evaluation

A. Latency and Pipeline Performance

We measured execution times during our empirical evaluation across 50

sample queries covering all intents. On average, the end-to-end journey took 2.7 seconds (σ = 0.6s). Inference latency from the two LLM calls dominated the timeline, demanding roughly 0.9 seconds each. Arithmetic processing and Pandas indexing combined for less than 50 milliseconds.

Table I — Breakdowns of processing delay.

Pipeline Stage	Mean	Std Dev	Notes
Intent Agent	0.91s	0.18s	LLM inference, temp=0.1
Retrieval	0.03s	0.02s	Pandas masking
Analysis	0.01s	0.01s	Python arithmetic
Reasoning	0.94s	0.24s	LLM inference, temp=0.4
Response	0.01s	0.01s	JSON serialisation
API overhead	0.08s	0.04s	HTTP transfers
Overall	2.70s	0.60s	N=50 test cases

Clearly, inference speed is the bottleneck. The structured data passing is basically instantaneous at this scale.

B. Intent Classification Accuracy

Using a holdout set of 140 queries, we tested the Intent Agent's precision. It achieved a 93.6% accuracy rate overall. The few errors that occurred were strictly misclassifications between adjacent topics, such as confusing a rent_query with an affordability question. Positively, it never mistook an off-topic question for a housing query.

Entity extraction worked well too. County targeting hovered around 97.1%, while year extraction hit 98.6%. Bedroom extraction was slightly lower at 94.3%, mostly failing when users used vague slang.

C. Ablation: ASD-RAG vs. Direct LLM Generation

We ran 50 factual queries against three system setups: direct LLM generation, ASD-RAG without the Analysis Agent, and the full pipeline. External annotators checked every generated figure against the actual CSV records.

Table II — Results from the ablation experiment.

Configuration	Fact-Check Hit Rate	Accurate Sources
Direct LLM	38.2%	0.0%
Retrieval Only	91.4%	100.0%
Full Pipeline	99.1%	100.0%

The explicit constraints of our retrieval operations handle the heavy lifting for factual grounding. Adding the strict Python environments of the Analysis Agent closes the final accuracy gap. Without it, the LLM still occasionally fumbles basic mortgage arithmetic even when successfully holding the correct base property price in context. Furthermore, during our direct LLM tests, the models frequently suffered from localized semantic hallucination: for instance, confidently misapplying historical UK housing policy initiatives or outdated affordability schemes directly to the distinct Irish market. They also tended to conflate listing asking prices scraped from old web articles with actual recorded transaction prices from the PPR. These systemic failures powerfully underscore exactly why deterministic tabular retrieval mechanisms remain mandatory for reliable financial query resolution.

D. Guardrail Effectiveness

We evaluated the system against 40 purely off-topic queries covering sports, politics, and generic coding questions. The Reasoning Agent identified and refused 39 of them (97.5%). The single slip-up involved a question about "American mortgage rates." The model briefly attempted to answer before stopping. While system prompts defend against obvious domain breaks, they occasionally struggle with queries that run parallel to the intended topic.

V. Conclusions and Future Work

A. Summary

The proposed architecture proves that combining Agentic RAG with deterministic tabular filters is highly effective for government data. Processing requests in under 3 seconds, the current pipeline managed 93.6% classification accuracy and 99.1% factual reliability. Direct LLM approaches recorded only 38.2% accuracy in our empirical evaluation. Fusing strict Python math with natural language explanation creates a system that is both readable and auditable.

B. Limitations

There are several clear technical limitations restricting the current pipeline. First, the setup relies entirely on static, pre-downloaded CSV files, meaning any sudden macroeconomic market shifts or new government dataset releases won’t register within the intelligence system until the next manual codebase update. Second, to manage computational memory thresholds during prototyping, the property sale dataset strictly covers the Dublin municipality, thereby limiting deeper property price analysis to the capital matrix. Additionally, without persistent conversational

session memory, users cannot engage in organic dialog follow-ups. If a user queries average Dublin rents, and subsequently asks "how does that compare to Galway?", the stateless LangGraph pipeline will currently fail to contextually resolve the foundational pronoun reference.

C. Future Work

Integrating live data APIs from the RTB and CSO portals poses the most practical next step. From there, implementing persistent state memory would allow for genuine dialogue chains. Moving the data to PostgreSQL PostGIS would open up neighborhood-level spatial analysis. Conducting a targeted user study to see if this improves decision-making in the real world would provide the most illuminating feedback.

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